

VERITAS SHIELD

AI-Native Consumer Defense Agent | Hackathon Proposal 2026

1. Project Vision

Veritas Shield is an AI-powered consumer advocate designed to protect individuals from "Dark Patterns," predatory contract clauses, and hidden subscription traps. In a world of complex legal jargon and deceptive digital interfaces, Veritas Shield acts as a real-time defense layer, using multimodal AI to interpret the world through a smartphone's eyes and ears.

2. Market Analysis & Competition

While tools exist for legal automation, Veritas Shield occupies a unique "AI-Native" niche by focusing on real-time mobility and multimodal detection.

Competitor	Focus	Limitation	Our Competitive Edge
DoNotPay	Legal Automation	Static forms; lacks real-time vision/context.	Multimodal Agency: Real-time detection of digital/physical traps.
ToS;DR	Web Summaries	Passive database; not personalized.	Active Shield: Analyzes <i>your</i> specific documents in <i>your</i> context.
Enterprise AI	Contract Review	Built for corporations; high cost.	Accessibility: Consumer-first UI designed for rapid mobile use.

3. Features & Architecture

The Mobile App ("The Shield")

- **Contract AR:** Point the camera at a document; the AI highlights "Red Flags" like auto-renewal clauses or hidden fees using AR overlays.

- **Dark Pattern Scanner:** Analyzes mobile UI/screenshots to identify "Deceptive Designs" (e.g., hidden cancel buttons).
- **Consent-Based Recording:** Summarizes verbal sales pitches to flag discrepancies between oral promises and written terms.

The Web Dashboard ("The Command Center")

- **Dispute Engine:** One-click generation of professionally drafted, legally cited dispute letters.
- **Risk Heatmap:** A community-driven map of companies with high rates of consumer complaints.

4. Technical Implementation & Challenges

Difficulty	Challenge	Proposed Solution
High	Legal Liability (UPL)	Frame as "Information Aggregator." Use RAG to cite specific laws, ensuring the AI isn't "interpreting" but "retrieving."
Med	OCR Accuracy	Combine Vision Transformers (ViT) with specialized layout analysis to preserve text hierarchy in messy documents.
Med	Agentic Reasoning	Utilize a multi-agent framework (e.g., LangChain) where one agent checks for "Tone" and another for "Legal Basis."

5. Team Roadmap (7-Day Sprint)

Day 1-2: Foundation

Finalize SRS/UML. AI Engineers setup RAG pipeline and Vision model baselines. UI/UX creates high-fidelity "Shield" mockups.

Day 3-4: Development

Backend integrates LLM API. Mobile Dev implements Camera/OCR pipeline. Frontend builds the Dispute Dashboard.

Day 5: Integration

Connect Mobile and Web via Backend. AI Engineers fine-tune the "Red Flag" detection logic and agentic output.

Day 6: Refinement

Stress testing and UI polish. Finalize the "Wow" demo scenario (e.g., canceling a high-profile subscription trap).

Day 7: Delivery

Record demo video, finalize documentation, and prepare the pitch for the committee.

Winning Strategy: Focus on "The First 30 Seconds." Ensure the judge sees a physical document scanned and a red flag detected immediately. Visual proof of AI agency is the key to first place.